

Physics-Informed Viscoelastic Plate Bending for Ice-Shelf Inversion: Inferring Thickness, Rheology, and Sub-Shelf Melt Rates from Remote Sensing Time Series

B. Minchew

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Abstract

We develop a framework for inverting remotely sensed time series of ice-shelf tidal flexure — from satellite altimetry (ICESat-2) and synthetic aperture radar interferometry (InSAR) — for ice-shelf thickness $h(x, y)$, effective rheological parameters, and sub-ice-shelf basal melt rates. The forward model is a 2D viscoelastic thin plate on an elastic (buoyancy) foundation, driven by multi-constituent tidal forcing. We argue that a Newtonian viscosity linearized at the observable background strain rate, combined with a Burgers (or Andrade) transient creep model, is more physically appropriate than a steady-state power-law viscosity for the tidal bending problem, where the Deborah number $De \sim 1$ and the microstructure cannot reach steady state within a tidal cycle. The frequency-dependent amplitude and phase response of the viscoelastic plate to multiple tidal constituents provides independent constraints that break the thickness–viscosity degeneracy inherent in single-frequency observations. Melt rates are recovered from the continuity equation using the inferred thickness field and accumulation from regional climate models. We outline a synthetic validation strategy using both custom FEM solvers and COMSOL, and a real-data application using the Rutford Ice Stream / Filchner–Ronne Ice Shelf system observed by COSMO-SkyMed SAR [Minchew et al., 2017].

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1 Introduction

Ice shelves modulate the discharge of grounded ice into the ocean by providing resistive backstresses through lateral drag at their margins and basal drag at pinning points [Thomas, 1979, Gudmundsson, 2013]. The rate at which ice shelves thin — and thus weaken — depends critically on sub-shelf basal melt rates, which remain among the most poorly constrained parameters in ice-sheet models. Current estimates of basal melt rely on the mass continuity equation, which requires knowledge of ice-shelf thickness $h(x, y, t)$, surface velocity, and surface mass balance. Thickness is typically derived from surface elevation under the hydrostatic assumption, which introduces errors from spatial variations in firn density, marine ice accretion, and the breakdown of hydrostaticity near grounding zones [Rignot et al., 2013].

Tidal flexure of ice shelves offers an independent, physics-based route to thickness and rheology. The vertical deflection of an ice shelf under tidal forcing is governed by the flexure equation, whose solution depends on the flexural rigidity $D \propto h^3$ and the rheological properties of the ice. Observations of tidal flexure — from repeat-track satellite altimetry, GPS, and InSAR — thus encode information about h and the constitutive behavior. The challenge is to extract this information through a well-posed inverse problem.

This document outlines a framework for doing so. The key ideas are:

1. A 2D viscoelastic plate model with Newtonian viscosity (linearized at the background strain rate) and Burgers transient rheology, rather than a steady-state power-law.
2. Multi-frequency tidal forcing that provides independent spectral constraints on thickness and rheology.
3. Rheological constraints from the free-floating shelf (where the stress state is known) propagated downstream into the grounding zone.
4. Recovery of basal melt rates from the continuity equation using the inferred thickness field.

2 Forward Model: Viscoelastic Plate Flexure

2.1 The Maxwell Viscoelastic Flexure Equation

For an incompressible Newtonian viscous material ($\nu_v = 1/2$), the plane-stress and plane-strain formulations are identical, so the 1D beam and 2D plate flexure equations share the same structure. The Maxwell viscoelastic flexure equation for a thin plate floating on water of density ρ_w is

$$D_v \nabla^4 \dot{w} + \rho_w g w + t_r \rho_w g \dot{w} = q + t_r \dot{q}, \quad (1)$$

where $w(x, y, t)$ is the vertical deflection, $D_v = \eta h^3/3$ is the viscous flexural rigidity, $D = Eh^3/[12(1-\nu^2)]$ is the elastic flexural rigidity, and $t_r = D_v/D \approx 1.35 t_M$ is the flexural relaxation time [see IceRheologyPart2, Appendix F].

For spatially varying thickness, D_v and D depend on (x, y) and must remain inside the differential operators. The biharmonic operator ∇^4 is replaced by the full variable-coefficient plate operator:

$$\frac{\partial^2}{\partial x^2} \left[D_v \left(\frac{\partial^2 \dot{w}}{\partial x^2} + \frac{1}{2} \frac{\partial^2 \dot{w}}{\partial y^2} \right) \right] + 2 \frac{\partial^2}{\partial x \partial y} \left[\frac{D_v}{2} \frac{\partial^2 \dot{w}}{\partial x \partial y} \right] + \frac{\partial^2}{\partial y^2} \left[D_v \left(\frac{\partial^2 \dot{w}}{\partial y^2} + \frac{1}{2} \frac{\partial^2 \dot{w}}{\partial x^2} \right) \right] + \rho_w g w + t_r \rho_w g \dot{w} = q + t_r \dot{q}. \quad (2)$$

2.2 Why Not Steady-State Power-Law Viscosity?

The standard Glen flow law $\dot{\varepsilon}_{ij} = A \tau_e^{n-1} \tau_{ij}$ describes the steady-state creep rate after the dislocation structure (or grain-boundary-sliding mechanism) has equilibrated to the current stress. During tidal bending, the stress at any material point oscillates through zero twice per tidal cycle. The microstructure cannot reorganize on this timescale: the Deborah number $De = t_M/T_{\text{tide}} \sim 0.1$ –1 for warm ice near grounding zones, meaning the relaxation time and the forcing period are comparable.

In this regime, the instantaneous viscosity during bending is not the steady-state power-law viscosity at the instantaneous stress. It is closer to the viscosity set by the time-averaged (background) stress, with the tidal perturbation experiencing a linearized (Newtonian) response around that background state. This is consistent with the critical tidal amplitude analysis: when the tidal bending strain rate exceeds the background strain rate, the power-law nonlinearity is formally active, but this is precisely the regime where the steady-state assumption is most questionable.

A Newtonian viscosity $\eta = 1/(2A \tau_{\text{bg}}^{n-1})$, evaluated at the observable background effective stress τ_{bg} , combined with a transient creep model, provides a more physically grounded description.

2.3 Burgers Transient Rheology

The Burgers model places a Maxwell element (spring μ_1 + dashpot η_1) in series with a Kelvin–Voigt element (spring μ_2 + dashpot η_2). The total strain rate is

$$\dot{\varepsilon}_{ij} = \frac{\dot{\tau}_{ij}}{2\mu_1} + \frac{\tau_{ij}}{2\eta_1} + \frac{\tau_{ij} - 2\mu_2 \varepsilon_{ij}^{(KV)}}{2\eta_2}, \quad (3)$$

where $\varepsilon_{ij}^{(KV)}$ is the strain in the Kelvin–Voigt element, governed by $\dot{\varepsilon}_{ij}^{(KV)} = (\tau_{ij} - 2\mu_2 \varepsilon_{ij}^{(KV)})/(2\eta_2)$. The Burgers model has two characteristic timescales:

- $t_M = \eta_1/\mu_1$: the Maxwell relaxation time (long-term viscous flow).
- $t_K = \eta_2/\mu_2$: the Kelvin–Voigt retardation time (transient creep).

When $t_K \ll T_{\text{tide}} \ll t_M$, the Kelvin–Voigt element has fully relaxed and the response is Maxwell viscoelastic. When $t_K \sim T_{\text{tide}}$, the transient element stiffens the response relative to the Maxwell prediction — the material is “stiffer than it looks” at tidal frequencies.

Under harmonic forcing at frequency ω , the Burgers model has a complex compliance

$$J^*(\omega) = \frac{1}{\mu_1} \left(\frac{1}{1 + i\omega t_M} \right) + \frac{1}{\mu_2} \left(\frac{1}{1 + i\omega t_K} \right), \quad (4)$$

which produces a frequency-dependent flexural response. Multi-constituent tidal forcing (M_2 , S_2 , O_1 , K_1 , M_{sf}) probes this compliance at multiple frequencies, providing independent constraints on t_M and t_K (and hence η_1 , μ_2 , η_2 , given that μ_1 is well-constrained from seismic measurements).

2.4 Harmonic Response of the Viscoelastic Plate

For monochromatic tidal forcing $q = \rho_w g w_0 \sin(\omega t)$, the steady-state deflection $w(x, y, t) = \text{Im}[W(x, y) e^{i\omega t}]$ satisfies

$$i\omega D_v \nabla^4 W + \rho_w g (1 + i\omega t_r) W = \rho_w g (1 + i\omega t_r) w_0. \quad (5)$$

The complex amplitude $W(x, y)$ encodes both the spatial amplitude $|W|$ and phase lag $\varphi = -\arg W$ of the tidal response.

For the Burgers model, t_r is replaced by a frequency-dependent complex relaxation time derived from the complex compliance $J^*(\omega)$. The key point is that at each tidal frequency ω_i , the plate has a different effective relaxation time, producing a different amplitude and phase pattern. Observing $|W|$ and φ at multiple frequencies overconstrains the system relative to a single-frequency observation.

3 Inverse Problem

3.1 Observables

The primary observables are:

- **Vertical tidal displacement** $w(x, y, t)$ from ICESat-2 repeat-track altimetry or InSAR. Amplitudes and phases at each tidal constituent are extracted by least-squares fitting to the displacement time series [Minchew et al., 2017].
- **Secular horizontal velocity** $\mathbf{v}(x, y)$ from InSAR or feature tracking, providing the background strain-rate field $\dot{\epsilon}_{\text{bg}}(x, y)$.
- **Horizontal tidal velocity variations** at the M_{sf} and other periods, which constrain the upstream propagation of tidal stresses and the basal mechanics [Minchew et al., 2017, Gudmundsson, 2006, Rosier et al., 2014].

3.2 Parameters to Invert For

1. **Ice thickness** $h(x, y)$. Enters through $D \propto h^3$ (elastic) and $D_v \propto h^3$ (viscous), so the flexure pattern is highly sensitive to thickness.
2. **Effective Newtonian viscosity** $\eta(x, y)$. Constrained independently on the free-floating shelf from the secular velocity field (where the stress state is known from the membrane stress balance), then propagated downstream into the grounding zone.
3. **Transient rheological parameters** t_K (and μ_2/μ_1). Constrained by the frequency dependence of the tidal response.
4. **Basal melt rate** $\dot{m}(x, y)$. Not a parameter of the flexure model, but recovered from the mass continuity equation once $h(x, y)$ is known:

$$\frac{\partial h}{\partial t} + \nabla \cdot (h \mathbf{v}) = \dot{a}_s - \dot{m}, \quad (6)$$

where $\dot{a}_s$ is the surface accumulation rate from regional climate models.

3.3 Breaking the Thickness–Viscosity Degeneracy

The flexural rigidity $D_v = \eta h^3/3$ means that thickness and viscosity trade off in the amplitude of the tidal response. Three strategies break this degeneracy:

1. **Phase information.** The phase lag φ depends on the ratio $\omega t_r = \omega D_v/D$. Since $D_v/D = 4\eta(1-\nu^2)/E$ is independent of h (the h^3 factors cancel), the phase constrains η independently of h . The amplitude then determines h .
2. **Multi-frequency observations.** The frequency dependence of $|W(\omega)|$ and $\varphi(\omega)$ has a different functional form for thickness changes versus viscosity changes. The M₂ (12.4 h), O₁ (25.8 h), and M_{sf} (14.77 d) constituents probe the rheology at three very different frequencies, providing strong leverage.
3. **Upstream rheological constraints.** On the free-floating shelf far from the grounding zone, the ice is in membrane stress balance with known boundary conditions (ocean pressure at the calving front). The secular velocity field and the Glen flow law determine $\eta(x, y)$ on the shelf. This viscosity, advected downstream, constrains the viscosity in the grounding zone and reduces the inversion to primarily a thickness problem there.

3.4 Inversion Methodology

Following the Bayesian framework of Tarantola [2005] as applied in Minchew et al. [2017], we write the posterior estimate of the model parameters $\mathbf{m} = [h, \eta, t_K, \dots]$ given observations $\mathbf{d}$ as

$$\tilde{\mathbf{m}} = (\mathbf{G}^T \mathbf{C}_d^{-1} \mathbf{G} + \mathbf{C}_m^{-1})^{-1} (\mathbf{G}^T \mathbf{C}_d^{-1} \mathbf{d} + \mathbf{C}_m^{-1} \mathbf{m}_0), \quad (7)$$

where $\mathbf{G}$ is the linearized forward operator (sensitivity of tidal amplitudes and phases to the parameters), $\mathbf{C}_d$ is the data covariance, $\mathbf{C}_m$ is the prior model covariance, and $\mathbf{m}_0$ is the prior model.

For the nonlinear viscoelastic plate, $\mathbf{G}$ is computed by linearizing the forward model around the current parameter estimate, and the inversion is iterated (Gauss–Newton or similar). The prior $\mathbf{m}_0$ for thickness comes from the hydrostatic estimate; the prior for viscosity comes from the shelf-interior estimate; the prior for t_K is informed by laboratory experiments.

4 Controlled Experiments: Synthetic Validation

Before applying the inversion to real data, we validate it on synthetic observations generated from a known “true” model.

4.1 Synthetic Model Design

The synthetic ice shelf is designed to resemble the Rutford Ice Stream / FRIS grounding zone:

- Domain: ~ 200 km along-flow $\times$ ~ 30 km cross-flow, with a U-shaped grounding line.
- Thickness: h varies from ~ 2 km at the grounding line to ~ 500 m at the shelf front, with cross-flow variations reflecting the channel geometry.
- Rheology: Newtonian viscosity $\eta(x, y)$ set by the Glen flow law at the background strain rate, with spatial variations from temperature (cold interior, warm margins due to shear heating) and grain size.

- Tidal forcing: M_2 , S_2 , O_1 , K_1 constituents with realistic amplitudes (combined amplitude up to ~ 3 m near the Ronne Ice Shelf front).
- Burgers transient parameters: t_K chosen to produce measurable frequency dependence in the tidal response.

4.2 Forward Solvers

Two independent forward solvers generate the synthetic observations:

1. **Custom FEM solver.** Extension of the 1D Hermite-cubic FEM from the teaching code (`viscoelastic_beam_solvers.py`) to 2D using Bogner–Fox–Schmit rectangular elements (bicubic Hermite, tensor product of the 1D basis). This solver handles Newtonian viscoelasticity with variable thickness on a structured rectangular grid. It is fast enough for iterative inversion.
2. **COMSOL Multiphysics.** Full 3D viscoelastic simulation on the same geometry, using COMSOL’s Structural Mechanics module with a Burgers material model. This provides a “ground truth” that includes 3D stress effects near the grounding line that the thin-plate model cannot capture. Comparison of the plate-model and COMSOL solutions quantifies the plate-model bias and identifies regions where the thin-plate assumption breaks down.

4.3 Synthetic Observation Generation

From the forward solution $w(x, y, t)$, synthetic observations are generated following the methodology of Minchew et al. [2017]:

1. Compute the 3D displacement field $\mathbf{u}(\mathbf{r}, t) = \mathbf{v}'t + \sum_i [c_i \sin(\omega_i t + \phi_i)]$ at each tidal frequency (eq. 1 of Minchew et al. 2017).
2. Project onto SAR line-of-sight (LOS) vectors for realistic viewing geometries matching the COSMO-SkyMed 4-satellite constellation.
3. Form displacement pairs d_{ab} for all image pairs with $\Delta t < 10$ days.
4. Add Gaussian noise with variance estimated from InSAR coherence (eq. 18 of Minchew et al. 2017).
5. Invert the noisy synthetic displacements for amplitude and phase at each tidal constituent using the Bayesian method (eq. 22 of Minchew et al. 2017).

The recovered amplitude and phase maps are the inputs to the physics-informed plate inversion.

4.4 Inversion Tests

Key tests:

1. **Single-frequency recovery:** invert M_2 alone for h and η . Expect the thickness–viscosity degeneracy to limit accuracy.
2. **Multi-frequency recovery:** jointly invert $M_2 + O_1 + M_{sf}$. Expect substantially improved separation of h and η .

3. **Transient rheology detection:** test whether the Burgers t_K is recoverable from the frequency dependence.
4. **Upstream rheological constraint:** fix η on the free-floating shelf from the secular velocity field and invert only for h in the grounding zone. Quantify the improvement.
5. **Melt rate recovery:** compute melt rates from the inferred h via continuity and compare to the true melt rates.
6. **Plate-model bias:** compare plate-model inversions of COMSOL synthetics to the true parameters. Quantify the 3D-stress bias near the grounding line.

5 Application: Rutford Ice Stream

Rutford Ice Stream (RIS) is a natural laboratory for this problem. It has:

- Strong, well-characterized tidal forcing with combined amplitudes exceeding 3 m [Padman et al., 2002, Han et al., 2005].
- Dense spatiotemporal SAR observations from the COSMO-SkyMed (CSK) 4-satellite constellation, providing ~ 9 months of data with 1–8 day revisit times [Minchow et al., 2017].
- Well-constrained tidal constituents: vertical motion dominated by M_2 and O_1 at semidiurnal and diurnal periods; horizontal flow dominated by M_{sf} at the fortnightly period [Gudmundsson, 2006, Murray et al., 2007, Minchow et al., 2017].
- Known ice thickness and bed topography from radar sounding [Fretwell et al., 2013, King et al., 2016].
- Relatively stable flow over the observational period, allowing the displacement to be decomposed into secular velocity plus sinusoidal tidal components [Minchow et al., 2017].

5.1 Available Data

Minchow et al. [2017] inferred time-dependent 3D velocity fields from 1644 coherent CSK displacement fields along 32 unique satellite tracks. The posterior model at each grid point consists of:

- A 3D secular velocity vector.
- Amplitude and phase of 3D sinusoidal velocity variations at M_2 (12.4 h), O_1 (25.8 h), and M_{sf} (14.77 d) periods.

Key observations relevant to the inversion:

1. **Vertical M_2 and O_1 :** spatially homogeneous amplitude and phase over the central ice shelf trunk, with rapid amplitude decrease and phase lag increase in the grounding zone (Figures 7a–d of Minchow et al. 2017). The phase lag within the grounding zone (< 20 min for M_2) directly constrains the viscoelastic rheology.
2. **Along-flow M_{sf} :** amplitude highest near the grounding zone, propagating upstream at ~ 24 – 34 km/day and damping at ~ 2.6 mm/km (Figure 9 of Minchow et al. 2017). Sensitive to ice thickness and basal mechanics.
3. **Lateral shear strain rates:** reveal margin structure and periodic grounding effects (Figure 10 of Minchow et al. 2017).

5.2 Rheological Constraints from the Shelf Interior

On the free-floating shelf, the depth-averaged longitudinal and lateral stress balance is

$$\frac{\partial}{\partial x}[h(2\tau_{xx} + \tau_{yy})] + \frac{\partial}{\partial y}(h\tau_{xy}) = \tau_d, \quad (8)$$

where $\tau_d = -\rho_i g h \partial h / \partial x$ is the gravitational driving stress. The deviatoric stresses are related to the observed strain rates through the viscosity: $\tau_{ij} = 2\eta \dot{\epsilon}_{ij}$. With the velocity field from InSAR and the thickness from hydrostatic inversion (or from the flexure inversion itself, iteratively), the effective viscosity $\eta(x, y)$ on the shelf is determined.

This shelf-interior viscosity serves as a strong prior for the grounding-zone inversion. The viscosity in the grounding zone is inherited from upstream, modified by:

- Temperature changes (slow, constrained by the advection timescale).
- Shear heating in the margins [Perol and Rice, 2015, Suckale et al., 2014].
- Grain-size evolution (slow).

These variations are smooth and can be parameterized with a small number of degrees of freedom.

5.3 Inversion Strategy for RIS

1. Estimate $\eta(x, y)$ on the shelf from the secular velocity field and the membrane stress balance.
2. Use the M_2 and O_1 phase lag maps in the grounding zone to refine η (or equivalently t_M) locally, breaking the thickness–viscosity degeneracy.
3. Invert the M_2 and O_1 amplitude maps for $h(x, y)$ in the grounding zone, using η from step 2 as a fixed (or weakly varying) parameter.
4. Use the M_{sf} along-flow amplitudes and phases to constrain basal mechanics upstream of the grounding zone (complementary to, but distinct from, the flexure inversion).
5. Compute melt rates from the continuity equation using the inferred $h(x, y, t)$, the observed velocity field, and accumulation from RACMO or similar.
6. Validate against independent thickness estimates from radar sounding [Fretwell et al., 2013, King et al., 2016].

6 Summary and Next Steps

The key contributions of this framework are:

1. A physically motivated constitutive model (Newtonian + Burgers transient) for tidal bending, replacing the steady-state power-law which is not justified at $De \sim 1$.
2. Multi-frequency tidal constraints that break the thickness–viscosity degeneracy.
3. Upstream rheological constraints from the free-floating shelf that reduce the dimensionality of the grounding-zone inversion.
4. A clear path from flexure-derived thickness to basal melt rates via continuity.

Immediate next steps:

- Implement the 2D Bogner–Fox–Schmit FEM plate solver with Newtonian + Burgers rheology.
- Build the RIS-like synthetic geometry and generate COMSOL “ground truth” solutions.
- Implement the linearized inversion and test on synthetics.
- Apply to the Minchew et al. [2017] CSK data.

References

- Fretwell, P., et al. (2013), Bedmap2: Improved ice bed, surface and thickness datasets for Antarctica, *Cryosphere*, 7(1), 375–393, doi:10.5194/tc-7-375-2013.
- Gudmundsson, G. H. (2006), Fortnightly variations in the flow velocity of Rutford Ice Stream, West Antarctica, *Nature*, 444(7122), 1063–1064, doi:10.1038/nature05430.
- Gudmundsson, G. H., J. Krug, G. Durand, L. Favier, and O. Gagliardini (2012), The stability of grounding lines on retrograde slopes, *Cryosphere*, 6(6), 1497–1505, doi:10.5194/tc-6-1497-2012.
- Han, S. C., C. K. Shum, and K. Matsumoto (2005), GRACE observations of M2 and S2 ocean tides underneath the Filchner–Ronne and Larsen ice shelves, Antarctica, *Geophys. Res. Lett.*, 32, L20311, doi:10.1029/2005GL024296.
- Minchew, B. (2026), Ice Rheology Part 2: Elasticity and Viscoelasticity, Lecture notes, Ge 193 — Glacier Dynamics, MIT.
- King, E. C., H. D. Pritchard, and A. M. Smith (2016), Subglacial landforms beneath Rutford Ice Stream, Antarctica: Detailed bed topography from ice-penetrating radar, *Earth Syst. Sci. Data*, 8(1), 151–158, doi:10.5194/essd-8-151-2016.
- Minchew, B. M., M. Simons, B. Riel, and P. Milillo (2017), Tidally induced variations in vertical and horizontal motion on Rutford Ice Stream, West Antarctica, inferred from remotely sensed observations, *J. Geophys. Res. Earth Surf.*, 122, 167–190, doi:10.1002/2016JF003971.
- Murray, T., A. M. Smith, M. A. King, and G. P. Weedon (2007), Ice flow modulated by tides at up to annual periods at Rutford Ice Stream, West Antarctica, *Geophys. Res. Lett.*, 34, L18503, doi:10.1029/2007GL031207.
- Padman, L., H. A. Fricker, R. Coleman, S. Howard, and L. Erofeeva (2002), A new tide model for the Antarctic ice shelves and seas, *Ann. Glaciol.*, 34(1), 247–254, doi:10.3189/172756402781817752.
- Perol, T., and J. R. Rice (2015), Shear heating and weakening of the margins of West Antarctic ice streams, *Geophys. Res. Lett.*, 42, 3406–3413, doi:10.1002/2015GL063638.
- Rignot, E., S. Jacobs, J. Mouginot, and B. Scheuchl (2013), Ice-shelf melting around Antarctica, *Science*, 341(6143), 266–270, doi:10.1126/science.1235798.
- Rosier, S. H. R., G. H. Gudmundsson, and J. A. M. Green (2014), Insights into ice stream dynamics through modelling their response to tidal forcing, *Cryosphere*, 8(5), 1763–1775, doi:10.5194/tc-8-1763-2014.

- Suckale, J., J. D. Platt, T. Perol, and J. R. Rice (2014), Deformation-induced melting in the margins of the West Antarctic ice streams, *J. Geophys. Res. Earth Surf.*, 119, 1004–1025, doi:10.1002/2013JF003008.
- Tarantola, A. (2005), *Inverse Problem Theory and Methods for Model Parameter Estimation*, Soc. for Ind. and Appl. Math., Philadelphia, Pa.
- Thomas, R. (1979), The dynamics of marine ice sheets, *J. Glaciol.*, 24(90), 167–177.
- Thompson, J., M. Simons, and V. C. Tsai (2014), Modeling the elastic transmission of tidal stresses to great distances inland in channelized ice streams, *Cryosphere*, 8(6), 2007–2029, doi:10.5194/tc-8-2007-2014.